Numerical Ordinary Differential Equation

Euler's Method

Exercise1: Use Euler's method to solve the initial value problem

$$\frac{dy}{dt} = \frac{t - y}{2}$$

on $0 \leq t \leq 1$ with $h = 0.25$. The initial condition at $t = 0$ is $y = 1$.

Solution:

Euler's Method

Exercise2: Use Euler's method to numerically integrate

$$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$$

from $x = 0$ *to* $x = 4$ with a step size of 0.5. The initial condition at $x = 0$ is $y = 1$. The exact solution is given by

$$y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x + 1$$

Solution:

Euler's Method

Exercise3:

1) Use Euler's method to numerically integrate

$$\frac{dy}{dx} = 3y^2 - 5x - 1, \qquad y(0) = 0$$

from $x = 0$ to $x = 1$ with a step size of 0.25.

2) Use Euler's method to numerically integrate

$$\frac{dy}{dx} = 1 - \sin x^2, \qquad y(0) = 1$$

from $x = 0$ to $x = 2\pi$ with a step size of $\frac{\pi}{2}$.

2nd Order Runge-Kutta Methods

Exercise1: Use the Heun method, midpoint method and Ralston's method to numerically integrate

$$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$$

from $x = 0$ *to* $x = 4$ with a step size of 0.5. The initial condition at $x = 0$ is $y = 1$. The exact solution is given by

$$y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x + 1$$

Heun Method ($a_2 = \frac{1}{2}$)

Then, $a_1 = \frac{1}{2}$, and $p_1 = q_{11} = 1$

$$y_{i+1} = y_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$$

With,

$$k_1 = f(x_i, y_i)$$
$$k_2 = f(x_i + h, y_i + k_1 h)$$

Midpoint Method ($a_2 = 1$)

Then, $a_1 = 0$, and $p_1 = q_{11} = \frac{1}{2}$

$$y_{i+1} = y_i + k_2 h$$

With,

$$k_1 = f(x_i, y_i)$$
$$k_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h\right)$$

Ralston's method ($a_2 = \frac{2}{3}$)

Then, $a_1 = \frac{1}{3}$, and $p_1 = q_{11} = \frac{3}{4}$

$$y_{i+1} = y_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right)h$$

With,

$$k_1 = f(x_i, y_i)$$
$$k_2 = f\left(x_i + \frac{3}{4}h, y_i + \frac{3}{4}k_1 h\right)$$

2nd Order Runge-Kutta Methods

Exercise2: Use the second order Runge-Kutta method (Ralston's method) to numerically integrate

$$y' - \cos 2x - \sin 3y = 0, \qquad y(0) = 1$$

from $x = 0$ to $x = 1$ with a step size of 0.2.

2nd Order Runge-Kutta Methods

Exercise3: Use the second order Runge-Kutta method (Ralston's mehod) to numerically integrate

$$\frac{dy}{dx} + y = \sin x, \qquad y(0) = 1$$

from $x = 0$ to $x = 3$ with a step size of $h = 1$.

2nd Order Runge-Kutta Methods

Exercise4: Use the second order Runge-Kutta method (Ralston's mehod) to numerically integrate

$$\frac{dy}{dx} + x - y = 1, \qquad y(0) = -1$$

from $x = 0$ to $x = 2$ with a step size of $h = 0.5$. Given the exact solution is $y = x - e^x$, find the true percent relative error of $y(2)$.

[Hint: True percent relative error=$\left|\frac{\text{true value} - \text{approximate value}}{\text{true value}}\right| \times 100\%$]